# 将 FlexVol 卷转换为 FlexGroup 卷

ONTAP 9

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# 将 FlexVol 卷转换为 FlexGroup 卷

## 将 FlexVol 卷转换为 FlexGroup 卷概述

如果要将 FlexVol 卷扩展到超出其空间限制的范围，则可以将 FlexVol 卷转换为 FlexGroup 卷。从 ONTAP 9.7 开始，您可以将具有 SnapMirror 关系的独立 FlexVol 卷或 FlexVol 卷转换为 FlexGroup 卷。

### 将 FlexVol 卷转换为 FlexGroup 卷的注意事项

在决定将 FlexVol 卷转换为 FlexGroup 卷之前，您应了解支持的功能和操作。

#### 转换期间不支持操作

在进行卷转换时，不允许执行以下操作：

- 卷移动
- 聚合自动执行
- 聚合重新定位
- 在高可用性配置中执行计划内接管和交还
- 在高可用性配置中手动和自动交还
- 集群升级和还原
- FlexClone 卷拆分
- 卷重新托管
- 卷修改和自动调整大小
- 卷重命名
- 将对象存储附加到聚合
- MetroCluster 配置中的协商切换
- SnapMirror 操作
- 从 Snapshot 副本还原
- 配额操作
- 存储效率操作

成功转换后，您可以对 FlexGroup 卷执行这些操作。

#### FlexGroup 卷不支持的配置

- 使卷脱机或受限
- SVM 根卷
- SnapLock 卷

- SAN
- SMB 1.0
- NVMe 命名空间
- 远程卷影复制服务（VSS）

# 将 FlexVol 卷转换为 FlexGroup 卷

从 ONTAP 9.7 开始，您可以将 FlexVol 卷原位转换为 FlexGroup 卷，而无需数据副本或额外的磁盘空间。

您需要什么？ **#8217**；将需要什么

- 过渡后的卷可以转换为 FlexGroup 卷。

  要将过渡后的卷转换为 FlexGroup 卷，需要使用 diag 模式，并且只能在 NetApp 支持部门的监督下执行。强烈建议您在继续进行转换过程之前联系 NetApp 支持部门。

- 要转换的 FlexVol 卷必须处于联机状态。
- FlexVol 卷上的操作和配置必须与转换过程兼容。

  如果 FlexVol 卷不兼容，并且卷转换已中止，则会生成一条错误消息。您可以采取更正操作并重试转换。

步骤

1. 验证 FlexVol 卷是否联机： `volume show vol_name -volume-style-Extended ， state`

```
cluster-1::> volume show my_volume -fields volume-style-extended,state
vserver volume    state  volume-style-extended
------- --------- ------ ---------------------
vs0     my_volume online flexvol
```

2. 验证是否可以无问题地转换 FlexVol 卷：

   a. 登录到高级权限模式： `set -privilege advanced`

   b. 验证转换过程： `volume conversion start -vserver vs1 -volume FlexVol -check -only true`

   在转换卷之前，必须更正所有错误。

   > 您不能将 FlexGroup 卷转换回 FlexVol 卷。如果要执行此操作，应联系支持部门。

3. 开始转换： `volume conversion start -vserver svm_name -volume vol_name`

```
cluster-1::*> volume conversion start -vserver vs0 -volume my_volume

Warning: Converting flexible volume "my_volume" in Vserver "vs0" to a
FlexGroup
         will cause the state of all Snapshot copies from the volume to
be set
         to "pre-conversion". Pre-conversion Snapshot copies cannot be
         restored.
Do you want to continue? {y|n}: y
[Job 57] Job succeeded: success
```

4. 验证转换是否成功： `volume show vol_name -fields -volume-style-Extended ， state`

```
cluster-1::*> volume show my_volume -fields volume-style-extended,state
vserver volume    state  volume-style-extended
------- --------- ------ ---------------------
vs0     my_volume online flexgroup
```

FlexVol 卷将转换为单成员 FlexGroup 卷。

您可以根据需要扩展 FlexGroup 卷。

# 将 FlexVol 卷 SnapMirror 关系转换为 FlexGroup 卷 SnapMirror 关系

## 要在 ONTAP 中将 FlexVol 卷 SnapMirror 关系转换为 FlexGroup 卷 SnapMirror 关系，必须先转换目标 FlexVol 卷，然后再转换源 FlexVol 卷。

您需要什么？ **#8217**；将需要什么

- 要转换的 FlexVol 卷必须处于联机状态。
- SnapMirror 关系中的源 FlexVol 卷不能是多个 SnapMirror 关系的源卷。

  FlexGroup 卷不支持扇出 SnapMirror 关系。

- FlexVol 卷上的操作和配置必须与转换过程兼容。

  如果 FlexVol 卷不兼容，并且卷转换已中止，则会生成一条错误消息。您可以采取更正操作并重试转换。

只有异步 SnapMirror 关系才支持 FlexGroup 转换。

步骤

1. 验证 SnapMirror 关系是否运行正常： `snapmirror show`

   只能转换 XDP 类型的镜像关系。

```
cluster2::> snapmirror show

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vs0:src_dpv DP   vs2:dst_dpv  Snapmirrored
                                      Idle           -         true    -
vs0:src_xdp XDP  vs2:dst_xdp  Snapmirrored
                                      Idle           -         true    -
```

2. 验证源卷是否与转换兼容：
   a. 登录到高级权限模式：`set -privilege advanced`
   b. 验证转换过程：`volume conversion start -vserver vs1 -volume src_vol -check -only true`

      在转换卷之前，必须更正所有错误。

3. 将目标 FlexVol 卷转换为 FlexGroup 卷。
   a. 暂停 FlexVol SnapMirror 关系：`snapmirror quiesce -destination-path dest_svm ：dest_volume`

      ```
      cluster2::> snapmirror quiesce -destination-path vs2:dst_xdp
      ```

   b. 开始转换：`volume conversion start -vserver dest_svm -volume dest_volume`

      ```
      cluster-1::> volume conversion start -vserver vs2 -volume dst_xdp

      Warning: After the volume is converted to a FlexGroup, it will not be
      possible
      to change it back to a flexible volume.
      Do you want to continue? {y|n}: y

      [Job 510] Job succeeded: SnapMirror destination volume "dst_xdp" has
      been successfully converted to a FlexGroup volume.
      You must now convert the relationship's source volume, "vs0:src_xdp",
      to a FlexGroup.
      Then, re-establish the SnapMirror relationship using the "snapmirror
      resync" command.
      ```

4. 将源 FlexVol 卷转换为 FlexGroup 卷： `volume conversion start -vserver src_svm_name -volume src_vol_name`

```
cluster-1::> volume conversion start -vserver vs0 -volume src_xdp

Warning: Converting flexible volume "src_xdp" in Vserver "vs0" to a
FlexGroup
         will cause the state of all Snapshot copies from the volume to
be set
         to "pre-conversion". Pre-conversion Snapshot copies cannot be
         restored.
Do you want to continue? {y|n}: y
[Job 57] Job succeeded: success
```

5. 重新同步此关系： `snapmirror resync -destination-path dest_svm_name ： dest_volume`

```
cluster2::> snapmirror resync -destination-path vs2:dst_xdp
```

您必须确保在扩展源 FlexGroup 卷以包含更多成分卷时，目标卷也会进行扩展。

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